Stage-Wise CBAM and Global–Local Fusion for Fabric Defect Detection and Assessment

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***Abstract*—Quality assessment through manual inspection is a challenging task in the textile industry due to texture variability in fabrics. Detecting and localizing subtle structural defects in fine and stochastic fabrics remains challenging for conventional inspection systems. To address this limitation, this letter proposes a stage-wise CBAM (Convolutional Block Attention Module) -augmented EfficientNetB0 architecture for automated fabric defect assessment. The model integrates Region-of-Interest (ROI) guidance with global–local feature fusion to enhance defect representation. The CBAM is strategically embedded after stage 3, 5, and 7 of EfficientNetB0 to refine spatial and channel features. This stage-wise insertion suppresses the intrinsic texture variance and enhances defect saliency. An ROI-driven feature refinement mechanism is applied after global–local feature fusion to focus on defect-relevant regions. Cohen's Kappa score and 5-fold cross validation validate the model's reliability. The global contextual descriptors, combined with localized defect cues, predict the defect class, while geometric dimensions are obtained from COCO annotations. These annotated dimensions are mapped to the ASTM D5430 four-point inspection system which gives the severity grading.**

***Index Terms*—Fabric defect detection, EfficientNetB0, CBAM, GLNet, ASTM scoring.**

I. Introduction

THE textile industry plays a vital role in the global manufacturing economy where quality is paramount, with reduction in wastage during manufacturing. Industry estimates indicates that the textile defect incurs 45% to 65% of manufacturing losses [1]. Also, traditionally, fabric quality assessment has relied heavily on manual visual inspection, which is labor intensive and subjective in nature. Human inspectors typically achieve a defect detection accuracy of merely 60% to 75% [1], [2]. Early automated methods depended on handcrafted descriptors derived from the image statistics. Classical statistical texture analysis approaches uses GLCM, Gabor filter, Fourier transform analysis and morphological operators were applied to analyze binary or grayscale fabric images [3]. While these methods are simple under controlled illumination, they suffer from sensitivity to texture variation and have limited generalization across the defect categories. The deep learning models show performance gains over handcrafted feature-based baselines. Recently, researchers have developed many approaches including one-stage detectors, two-stage region-based detectors, segmentation networks, and unsupervised anomaly detection frameworks [4]. Table I provides a comparison of recent methods based on architecture and functional capabilities. These methods either focus on attention or localization, lacking integration of multi-stage attention, global–local feature fusion, and ROI-guided refinement. This gap forms the basis of our proposed method.

To bridge these gaps, we propose a model that combines four deep learning components. EfficientNetB0 is selected as the backbone architecture. Then CBAM is embedded after 3rd, 5th and 7th stage of the backbone to suppress irrelevant textural noise in both channel and spatial dimensions. GLNet performs global–local feature fusion, followed by ROI-guided attention to enhance defect localization.

This paper is structured as follows. Section II reviews related work in the domain. Section III describes the dataset used and proposed methodology in detail. Section IV presents the experimental setup, results, an ablation analysis, and Section V conclusion.

II. Related Work

Fabric defect detection has largely depended on CNN-based methods. Early work, such as region-membership-degree–based CNN [5], brought spatial statistics into learned representations. Later, detection and segmentation improved through frameworks like RefineDet [6], multi-scale CNNs [7], and lightweight architectures such as Mobile-UNet [8]. Hybrid models combined handcrafted features with deep learning models, including Gabor-enhanced Faster R-CNN [9]. Attention mechanisms are also widely adopted to enhance feature discrimination [10], like integrating CBAM with EfficientNet and YOLO-based detectors [11] and region-based multi-scale training strategies [12]. Some recent works explore frequency-domain features [17]; however, they do not integrate stage-wise Global-Local feature fusion with ROI-based localization and automated quality scoring.

III. Methodology

The proposed model is trained using images from the TILDA fabric dataset. The dataset contains fabric images belonging to two structural classes. Class 1 includes fabrics without visible internal structure (e.g., viscose or silk), and Class 2 contains fabrics with stochastic structures of low variance (e.g., wool or jute). The dataset consists of five defect categories (E0-Defect Free, E1-Hole, E2-Stain, E3–Thread, E4-Foreign-Body), each category contains 200 images (100 per class), resulting in 1000 images. The original images are grayscale .tiff files with a resolution of 768 $\times$ 512 pixels and 8 $\times$ 30 DPI, corresponding

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TABLE I
COMPARATIVE ANALYSIS OF EXISTING FABRIC DEFECT DETECTION METHODS AND THE PROPOSED APPROACH

Reference	Method / Architecture	Attention mechanism	ROI / localization	GLNet	Quality scoring	Dataset
[11] Dai et al., 2025	CBAM-EfficientNet + YOLO	✔	✗	✗	✗	DAGM2007
[13] Hassan et al., 2024	Enhanced DCNN + HRI	✓	✗	✗	✗	Custom
[14] Zhou et al., 2025	DCFE-YOLO (YOLOv8 + DSConv)	✓	✓	✗	✗	TILDA / Custom
[15] Chen et al., 2025	YOLOv8 + Swin-T + BiFPN + CBAM	✓	✓	✗	✗	Tianchi
[16] Lin et al., 2024	Fabric4Show (Faster R-CNN)	✗	✓	✗	✓	Tianchi /
[17] Gu et al., 2025	DF-YOLO (Fourier + BiFPN + CBAM)	✓	✗	✗	✗	TILDA
[18] Jin et al., 2025	Improved YOLOv8 (RG-C2f + DySample)	✓	✓	✗	✗	TILDA / Tianchi
[19] Wang et al., 2025	Fab-ME (YOLOv8s + VMamba + EMCA)	✓	✓	✗	✗	TILDA / Tianchi
Proposed	**EfficientNetB0 + CBAM + ROI + GLNet**	✓	✓	✓	✓	**TILDA**

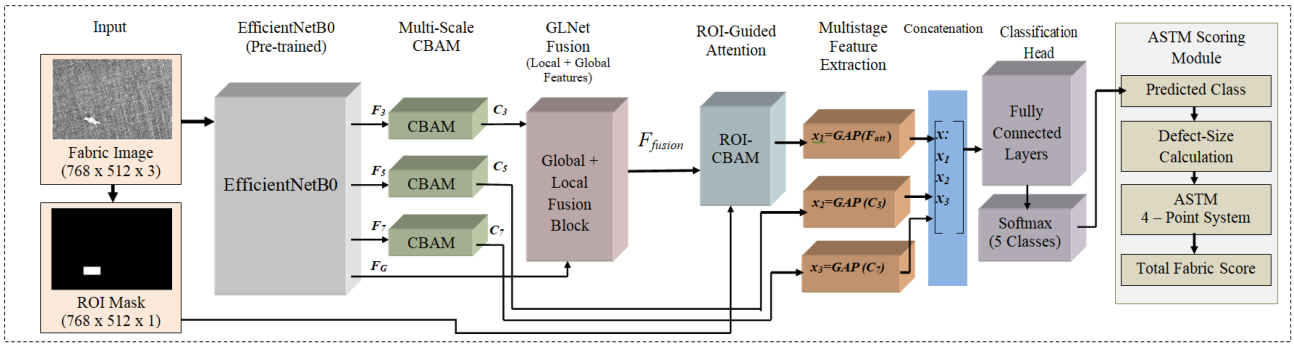


Fig. 1. Proposed Framework for fabric defect detection using EfficientNetB0, CBAM, Fusion, ROI Attention, and ASTM scoring.

TABLE II
PERFORMANCE COMPARISON OF BACKBONE NETWORKS

Backbone Model	Acc.	Precision	Recall	F1-Score	AUC
EfficientNetB0	**0.8100**	**0.8252**	**0.8387**	**0.8319**	**0.8968**
MobileNetV2	0.7800	0.7741	0.7988	0.7862	0.8571
ShuffleNetV2	0.7300	0.7263	0.7492	0.7376	0.8341

physical size is approximately 96 × 17.07 inches. For model training and evaluation the dataset was split into 80% training, 10% validation, and 10% testing. Images were converted to .jpg, annotated using Roboflow, and exported in COCO format for model training.

Initially the lightweight baseline models EfficientNetB0, MobileNetV2, and ShuffleNetV2 were trained and evaluated on the dataset; corresponding results are summarized in Table II.

Fig. 1 shows the proposed architecture, which integrates EfficientNetB0, stage-wise CBAM, Global-Local feature fusion, ROI-guided attention and, an ASTM inspired scoring module to evaluate fabric quality.

The methodology is described in the following subsections:

A. Problem Formulation

Let the input fabric image *(I)* with height *(H)* and width *(W)* be denoted as

$$\mathrm{I} \in \mathbb{R}^{\mathrm{H\ x\ W\ x\ 3}} \tag{1}$$

and the corresponding Region-of-Interest (ROI) mask *(M)* derived from COCO annotations as

$$\mathrm{M} \in \{0,1\}^{\mathrm{H\ x\ W}} \tag{2}$$

The objective is to learn a mapping $(f(.))$

$$f : (\mathrm{I}, \mathrm{M}) \rightarrow \mathrm{y} \in \mathbb{R}^{\mathrm{C}} \tag{3}$$

where $C = 5$ represents defect classes:

$C = \{Defect - Free, Hole, Thread, Stain, Foreign\ Body\}$

The framework also estimates an ASTM-inspired fabric quality score (S).

B. EfficientNetB0 Feature Extraction

Given the input defined in (1), hierarchical feature *(F)* extraction is performed using EfficientNetB0:

$$F = \phi\ (I) \tag{4}$$

Intermediate feature maps $(F_3,\ F_5, F_7)$ are extracted at three stages as per (4).

$$F_3 \in \mathbb{R}^{\mathrm{H_3\ x\ W_3\ x\ 24}},\ F_5 \in \mathbb{R}^{\mathrm{H_5\ x\ W_5\ x\ 80}},\\ F_7 \in \mathbb{R}^{\mathrm{H_7\ x\ W_7\ x\ 192}} \tag{5}$$

and the final global feature (F_g)

$$F_g \in \mathbb{R}^{\mathrm{H_g\ x\ W_g\ x\ 1280}} \tag{6}$$

C. Multi-Scale CBAM Attention

To enhance feature learning for each stage (5) feature map F_i, a Convolutional Block Attention Module (CBAM) is applied:

1) Channel Attention (A_C) is calculated as

$$A_C = \sigma(W_2.\delta(W_1.AvgPool(F_i)) \\ + W_2.\delta(W_1.MaxPool(F_i))) \tag{7}$$

The intermediate refined feature map F_i' is obtained

$$F_i' = F_i\ \odot A_C \tag{8}$$

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2) Spatial Attention (A_S) is calculated as

$$A_S = \sigma\left(f^{7\ x\ 7}\left(\left[AvgPool\left(F_i'\right), MaxPool\left(F_i'\right)\right]\right)\right) \quad (9)$$

where $f^{7\ x\ 7}$ denotes convolution operation with a 7 *x* 7 kernel. The final refined feature map (F_i'') is given as

$$F_i'' = F_i' \odot A_S \quad (10)$$

Thus, refined multi-scale features maps after CBAM are $C_3,\ C_5\ ,\ C_7$ are obtained.

D. Global-Local Feature Fusion (GLNet)

To integrate semantic and texture-level features, a Global-Local Fusion block is employed:

$$F_g' = Conv_{1\ x\ 1}\left(F_g\right),\ C_3' = Resize\left(C_3\right) \quad (11)$$

$$C_3'' = Conv_{1\ x\ 1}\ \left(C_3'\right) \quad (12)$$

where F_g' is a Processed global feature map and C_3' is a resized local feature map

Fusion (F_{fusion}) is performed as:

$$F_{fusion} = Conv_{3\ x\ 3} \times \left(F_g' + C_3''\right) \quad (13)$$

E. ROI-Guided Attention Mechanism

The fused feature map in (13) is modulated using the resized ROI mask *(M)* to match the spatial dimensions.

$$F_{roi} = F_{fusion}\ \ \odot \sigma\ \left(Resize\ \left(M\right)\right) \quad (14)$$

CBAM is then applied to Final attention-enhanced feature representation (F_{att})

$$F_{att} = CBAM\left(F_{roi}\right) \quad (15)$$

This enhances defect localization.

F. Multi-Scale Feature Aggregation

Global average pooling is applied to obtain feature vectors $(x_1,\ x_2, x_3)$

$$x_1 = GAP\left(F_{att}\right) \in \mathbb{R}^{1280} \quad (16)$$

$$x_2 = GAP\left(C_5\right) \in \mathbb{R}^{80} \quad (17)$$

$$x_3 = GAP\left(C_7\right) \in \mathbb{R}^{192} \quad (18)$$

Concatenated multi-scale feature vector (x) is obtained

$$x = \left[x_1; x_2\ ; x_3\right] \in \mathbb{R}^{1552} \quad (19)$$

G. Classification Head

The final prediction is obtained as

$$z = \delta\left(W_d x + b_d\right) \quad (20)$$

where z is the latent feature vector, W_d is weight matrix and b_d is bias vector, δ is ReLU activation function.

The final classification output (y) is derived using Softmax function

$$y = Softmax\left(W_0 z + b_0\right) \quad (21)$$

H. Loss Function

Categorical cross-entropy is used:

$$L = -\sum_{i=1}^{C} y_i \log(\widehat{y_i}) \quad (22)$$

To address class imbalance, weighted loss is applied:

$$L_{weighted} = -\sum\nolimits_{i=1}^{C} w_i y_i \log(\widehat{y_i}) \quad (23)$$

where C denotes the number of defect classes, y_i is the ground truth, $\hat{y}$ is the predicted probability, and w_i represents the weight assigned to each class based on its frequency.

I. ASTM-Based Fabric Quality Scoring

For ASTM -inspired fabric quality assessment the defect dimensions are computed from the annotated COCO bounding boxes. The w_p and h_p denote the width and height of the annotated bounding box in pixels. These dimensions are converted into inches using D_h(horizontal DPI) and D_v (vertical DPI).

$$w_{in} = \frac{w_p}{D_h}\ \ h_{in} = \frac{h_p}{D_v} \quad (24)$$

As per the ASTM four point inspection system the penalty points are assigned based on the maximum dimension therefore the maximum dimension d is determined as:

$$d = \max\left(w_{in}, h_{in}\right) \quad (25)$$

Using the maximum dimension d, ASTM penalty points (P_k) are assigned as below

$$P_k = \begin{cases} 1, & d \leq 3 \\ 2, & 3 < d \leq 6 \\ 3, & 6 < d \leq 9 \\ 4, & d > 9\ or\ hole > 1 \end{cases} \quad (26)$$

The total fabric quality score S for a given sample containing N_d defects is calculated as

$$S = \sum\nolimits_{k=1}^{N_d} P_k \quad (27)$$

J. Explainability Via Grad-CAM

To visualize defect regions and to provide transparency in the decision-making process, Grad-CAM is computed:

$$a_k = \frac{1}{Z}\sum\nolimits_i \sum\nolimits_j \frac{\partial \mathrm{y}^c}{\partial \mathrm{A}_{i\ j}^k} \quad (28)$$

where,a_k is weight of the kth feature map and $Z = H\ x\ W$

The final heat map is generated by applying the ReLU activation

$$L_{GradCAM}^c = ReLU\left(\sum\nolimits_k a_k A^k\right) \quad (29)$$

K. Cohen's Kappa Coefficient

Finally, to assess the reliability of the classification beyond mere accuracy, the Cohen's Kappa is calculated

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (30)$$

where p_0 is the observed agreement and p_e is the expected agreement by chance. The Equations given above are the mathematical foundation of the proposed model.

IV. Results

The lightweight backbone architectures EfficientNetB0, MobileNetV2, and ShuffleNetV2 were evaluated, and the comparative results are summarized in Table II.

From Table II, it is clear that, the EfficientNetB0 performs better and hence, selected as the backbone architecture. CBAM modules were then integrated at stages 3, 5, and 7. The fusion strategies are compared and results are tabulated in Table III.

Table III shows that GLNet achieved the better performance and was therefore selected as the fusion strategy. The M1 to M4 models are evaluated to assess the contribution of each

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TABLE III
FUSION STRATEGY COMPARISON (WITHOUT ROI-GUIDED ATTENTION)

Fusion Strategy	Accuracy	F1-Score	Cohen's Kappa
Attention-Gated Fusion	0.8300	0.8265	0.7875
Lightweight FPN	0.8700	0.8656	0.8375
GLNet	0.9100	0.9142	0.8934

TABLE IV
STUDY OF PROPOSED MODEL COMPONENTS

Model	Acc.	Precision	Recall	F1-Score	AUC	Kappa
M1	0.8100	0.8252	0.8387	0.8319	0.8968	0.7643
M2	0.8900	0.9073	0.8784	0.8926	0.9231	0.8664
M3	0.9100	0.9264	0.9023	0.9142	0.9547	0.8934
M4	**0.9300**	**0.9482**	**0.9271**	**0.9375**	**0.9752**	**0.9250**

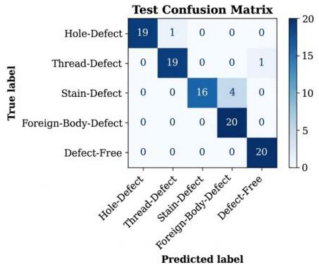


Fig. 3. Confusion matrix of the proposed model.

TABLE V
STRATIFIED 5-FOLD CROSS-VALIDATION RESULTS

Metric	Fold-1	Fold - 2	Fold-3	Fold-4	Fold-5	Mean ± Std
Accuracy	0.9250	0.9100	0.9300	0.9200	0.9441	0.9258 ± 0.0113
F1-Score	0.9241	0.9101	0.9307	0.9202	0.9457	0.9262 ± 0.0117
AUC	0.9821	0.9853	0.9877	0.9884	0.9976	0.9882 ± 0.0054

component in the proposed architecture. The M1 to M4 models are defined as follows:

- M1: EfficientNetB0 (baseline)
- M2: M1 + CBAM (Stages 3, 5, 7)
- M3: M2 + GLNet for feature fusion
- M4: M3 + ROI-guided attention (proposed model)

The above models and their findings are tabulated in Table IV. Table IV shows a consistent improvement in performance from M1 to M4.

The proposed model M4 (M3 + ROI guided attention) achieves the best overall performance, with an accuracy of 93.00%, and Cohen's Kappa score of 0.9250. This indicates that the proposed model has strong discriminative capability across all classes. The Global–Local Network enables the model to capture both fine-grained defect patterns and global fabric structure.

The confusion matrix shown in Fig. 3 gives the class-wise misclassification for test dataset. Table V shows the stratified 5-fold cross-validation results of the proposed model.

In ASTM-based fabric inspection the penalty score is assigned based on the maximum length of the defect. Defect dimensions are computed directly from the COCO-annotated bounding boxes, and corresponding penalty scores for sample test images

TABLE VI
ASTM-BASED DEFECT SCORING USING DEFECT DIMENSIONS

Image	Height (in)	Width (in)	Max Dimension (in)	Point (s)
Sample 1	2.70	6.62	6.62	3
Sample 2	2.45	8.25	8.25	3
Sample 3	13.75	2.16	13.75	4

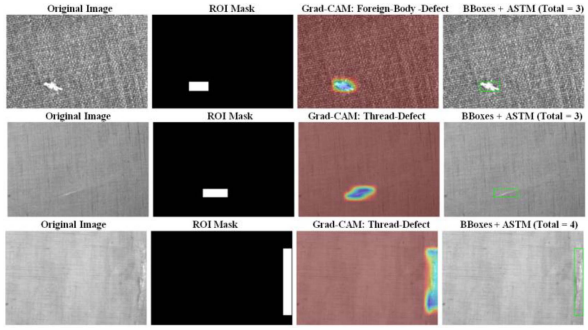


Fig. 2. Grad-CAM localization and ASTM penalty scores derived from COCO-annotated bounding boxes.

TABLE VII
CROSS-DATASET GENERALIZATION PERFORMANCE

Dataset	Resolution	Accuracy	F1-Score
Source Dataset			
TILDA	768 × 512	0.9300	0.9375
Target Datasets			
AITEX	4096 × 256	0.7623	0.7785
DAGM 2007	512 × 512	0.8254	0.8406
MVTec Carpet	1024 × 1024	0.8500	0.8944
MVTec Leather	1024 × 1024	0.9261	0.9172

are tabulated in Table VI. Fig. 2 shows the defective region along with the corresponding ASTM penalty points. Cross-dataset evaluation was performed on the target dataset using trained model with a consistent patch-based inference strategy and, the results are summarized in Table VII. The patch size is 768 × 512 pixel patches with 50% overlap. For AITEX, the image height was resized from 256 to 512 pixels before patch extraction.

The proposed model performed effectively across the target datasets without requiring retraining. The model performance on AITEX dataset reflects the effects of domain shift and image resizing.

V. CONCLUSION

The letter discussed above presented an EfficientNetB0-based architecture with CBAM attention mechanism, ROI-guided refinement of features and global-local fusion of features for fabric defects detection and their quantitative evaluation. The proposed architecture demonstrates 93.00% accuracy, AUC 0.9752, and a Cohen's Kappa value of 0.9250. The high Cohen's Kappa value and ablation study results demonstrate the reliability and robustness of the proposed architecture across different defect classes. Grad-CAM visualization and quantitative defect measurement further validate the effectiveness of the proposed framework.

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REFERENCES

[1] T. Ersöz, H. Zahoor, and F. Ersöz, "Fabric and production defect detection in the apparel industry using data mining algorithms," *Int. J. 3D Printing Technol. Digit. Ind.*, vol. 5, no. 3, pp. 742–757, Dec. 2021, doi: 10.46519/ij3dptdi.1030676.

[2] H. Y. T. Ngan, G. K. H. Pang, and N. H. C. Yung, "Automated fabric defect detection—A review," *Image Vis. Comput.*, vol. 29, no. 7, pp. 442–458, Jun. 2011, doi: 10.1016/j.imavis.2011.02.002.

[3] J. L. Raheja, S. Kumar, and A. Chaudhary, "Fabric defect detection based on GLCM and Gabor filter: A comparison," *Optik*, vol. 124, no. 23, pp. 6469–6474, Dec. 2013, doi: 10.1016/j.ijleo.2013.05.004.

[4] C. Li, J. Li, Y. Li, L. He, X. Fu, and J. Chen, "Fabric defect detection in textile manufacturing: A survey of the state of the art," *Secur. Commun. Netw.*, vol. 2021, 2021, Art. no. 9948808, doi: 10.1155/2021/9948808.

[5] L. Song, R. Li, and S. Chen, "Fabric defect detection based on membership degree of regions," *IEEE Access*, vol. 8, pp. 48752–48760, 2020, doi: 10.1109/ACCESS.2020.2978900.

[6] H. Xie and Z. Wu, "A robust fabric defect detection method based on improved RefineDet," *Sensors*, vol. 20, no. 15, Jul. 2020, Art. no. 4260, doi: 10.3390/s20154260.

[7] S. Mei, Y. Wang, and G. Wen, "Automatic fabric defect detection with a multi-scale convolutional denoising autoencoder network model," *Sensors*, vol. 18, no. 4, Apr. 2018, Art. no. 1064, doi: 10.3390/s18041064.

[8] J. Jing, Z. Wang, M. Rätsch, and H. Zhang, "Mobile-UNet: An efficient convolutional neural network for fabric defect detection," *Text Res. J.*, vol. 92, no. 1/2, pp. 30–42, Jan. 2022, doi: 10.1177/0040517520928604.

[9] M. Chen et al., "Improved faster R-CNN for fabric defect detection based on Gabor filter with genetic algorithm optimization," *Comput. Ind.*, vol. 134, 2022, Art. no. 103551, doi: 10.1016/j.compind.2021.103551.

[10] B. Fang, X. Long, F. Sun, H. Liu, S. Zhang, and C. Fang, "Tactile-based fabric defect detection using convolutional neural network with attention mechanism," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, Art. no. 5011309, doi: 10.1109/TIM.2022.3165254.

[11] N. Dai, X. Hu, K. Xu, X. Hu, Y. Yuan, and Y. Xu, "A study on lightweight algorithms for fabric defect detection," *Text Res. J.*, vol. 95, no. 21/22, 2025, Art. no. 00405175241293380, doi: 10.1177/00405175241293380.

[12] F. Li and F. Li, "Bag of tricks for fabric defect detection based on Cascade R-CNN," *Text Res. J.*, vol. 91, no. 5/6, pp. 599–612, Mar. 2021, doi: 10.1177/0040517520955229.

[13] S. A. Hassan, M. J. Beliatis, A. Radziwon, A. Menciassi, and C. M. Oddo, "Textile fabric defect detection using enhanced deep convolutional neural network with safe human–robot collaborative interaction," *Electronics*, vol. 13, no. 21, Nov. 2024, Art. no. 4314, doi: 10.3390/electronics13214314.

[14] L. Zhou, B. Ma, Y. Dong, Z. Yin, and F. Lu, "DCFE-YOLO: A novel fabric defect detection method," *Plos One*, vol. 20, no. 1, Jan. 2025, Art. no. e0314525, doi: 10.1371/journal.pone.0314525.

[15] C. Chen, Q. Zhou, S. Li, D. Luo, and G. Tan, "Fabric defect detection algorithm based on improved YOLOv8," *Text Res. J.*, vol. 95, no. 3/4, pp. 235–251, 2025, doi: 10.1177/00405175241261092.

[16] H. Lin, D. Cai, Z. Xu, J. Wu, L. Sun, and H. Jia, "Fabric4Show: Real-time vision system for fabric defect detection and post-processing," *Vis. Intell.*, vol. 2, no. 1, Jun. 2024, Art. no. 13, doi: 10.1007/s44267-024-00047-w.

[17] M. Gu, J. Zhou, R. Pan, and W. Gao, "DF-YOLO: An attempt on enhancing generalization in fabric defect detection based on YOLO network," *Text Res. J.*, vol. 95, no. 7/8, pp. 748–766, 2025, doi: 10.1177/00405175241269163.

[18] Y. Jin et al., "A real-time fabric defect detection method based on improved YOLOv8," *Appl. Sci.*, vol. 15, no. 6, 2025, Art. no. 3228, doi: 10.3390/app15063228.

[19] S. Wang, H. Kong, B. Li, and F. Zheng, "Fab-ME: A vision state-space and attention-enhanced framework for fabric defect detection," in *Proc. 21st Int. Conf. Intell. Comput.*, 2025, pp. 103–114, doi: 10.1007/978-981-96-9901-8_9.

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